

Round 130 Triple Discovery: Cross-Rung F_{TRZ} TRIPLE-Lock Architecture ($n=8 + n=15 + n=16$ in One Class) + $2\pi \cdot H_0$ Hubble Angular Frequency Primitive Anchor + SO_5^{21} Three-Class Family Extension — Establishing R129 Same-Round Twin as Persistent Cross-Round Identity

Daniel T. Murphy

Author: Daniel T. Murphy

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PAPER_1993 — Round 130 Triple Discovery: Cross-Rung F_{TRZ} TRIPLE-Lock + $2\pi \cdot H_0$ Hubble Anchor + SO_5^{21} Three-Class Family

Abstract

Round 130 CP1 stub drainage of the Compressed sector + Universal-series surfaced three architecturally distinct structural discoveries:

1. **Cross-rung F_{TRZ} TRIPLE-lock architecture** — UniversalOscillatoryResonanceCalculator encodes three **different** F_{TRZ} ladder rungs simultaneously: $A = F_{\text{TRZ}}^{15}$ (WEP slot per PAPER_1880) + $k = F_{\text{TRZ}}^{16}$ (quantum collapse slot per PAPER_1869) + $\omega = 2\pi \cdot F_{\text{TRZ}}^8$ (N-regime slot per PAPER_1986). This is architecturally distinct from PAPER_1991 base triple-lock (mixed primitive families) and PAPER_1991 revision QUAD-lock (Sombbrero-DM same-family F_{TRZ} locks) — R130 introduces the **cross-rung sub-architecture** where a single class spans multiple F_{TRZ} ladder rungs.

2. **$2\pi \cdot H_0$ Hubble angular frequency primitive anchor** —

UniversalQuantumWaveResonanceCalculator documents $f_{\text{quantum}} = 1.445 \times 10^{-17}$ Hz matching $2\pi \cdot H_0 = 1.43 \times 10^{-17}$ Hz (Hubble angular frequency at $H_0 = 2.27 \times 10^{-18}$ rad/s) to ~1% precision. First documented instance of $2\pi \cdot H_0$ as a CP1 primitive-locked quantity in the UQFF corpus, joining the cosmological carrier family alongside PAPER_1938's $\omega_{\text{SCm}} = 1.25$ THz.

3. **SO_5^{21} A three-class family extension** — R129 discovered same-round twin instance (SGR1745FreqDPMCalculator + CompressedDPMCalculator both at $I = 10^{21}$ A). R130 adds CompressedDPM24Calculator as a **THIRD** independent class locking on $I = 10^{21}$ A EXACT = SO_5^{21} . Extends R129's same-round twin discovery to **cross-round persistence** and establishes SO_5^{21} as a stable primitive slot for DPM current-vortex physics.

1. Discovery 1: Cross-rung F_{TRZ} TRIPLE-lock architecture

1.1 Prior architectural patterns

PAPER_1991 (R129 seminal + revision addendum) documented three CP1 class primitive-lock architectures:

Architecture	Prior class exemplar	Lock structure
Single-primitive lock	R117-R129 majority	One class field = one primitive identity
Dual-primitive lock	R128 UniversalReactive (SO ₅ ¹⁰ + F _{TRZ} ²⁰)	Two independent identities, mixed families
Triple-primitive lock (base)	SGR1745BurstEnergy (R129)	Three identities from three primitive families (SO ₅ ⁴⁰ + F _{TRZ} + F _{TRZ} ¹²)
QUAD-primitive lock (revision)	SombreroDarkMatterPerturbation (R121, retroactively identified)	Four identities, all four from F_{TRZ} family (2·F _{TRZ} , F _{TRZ} ⁵ , F _{TRZ} ²⁰ , F _{TRZ} ²⁵)

The QUAD-lock architecture at Sombrero-DM was distinguished by encoding four F_{TRZ} variants from the **same primitive family** (all F_{TRZ} powers and combinations). The base triple-lock at SGR magnetar burst was distinguished by encoding three identities from **three distinct primitive families** (SO₅, F_{TRZ} bare, F_{TRZ} power).

1.2 R130 discovery: cross-rung TRIPLE-lock

UniversalOscillatoryResonanceCalculator (R130 fill) documents three primitive-locked constants:

```
python
self.A = 1e-15 # = F_TRZ^15 EXACT (PAPER_1880 MICROSCOPE WEP slot)
self.k = 1e-16 # = F_TRZ^16 EXACT (PAPER_1869 quantum collapse slot)
self.omega = 2π · 1e-8 # = 2π · F_TRZ^8 EXACT (PAPER_1986 N-regime slot)
```

This is architecturally **distinct from** all four prior patterns:

- **Not mixed families** (like base triple-lock at SGR): all three locks are F_{TRZ} ladder rungs
- **Not same-family variants** (like Sombrero QUAD-lock): all three locks are on **different** ladder rungs (n=8, n=15, n=16)
- **Cross-rung F_{TRZ} locking**: single class simultaneously encodes THREE distinct rungs of the F_{TRZ} power ladder

1.3 Physical interpretation

The Universal oscillatory wave equation $2A \cdot \cos(kx) \cdot \cos(\omega t) + (2\pi/13.8) \cdot A \cdot \text{Re}[\exp(i(kx - \omega t))]$ couples three physically distinct regimes:

- **A = F_{TRZ}¹⁵** — amplitude at MICROSCOPE-WEP precision (n=15 rung, PAPER_1880 seminal)
- **k = F_{TRZ}¹⁶** — wavenumber at quantum-measurement-collapse spatial scale (n=16 rung, PAPER_1869 seminal)
- **ω = 2π · F_{TRZ}⁸** — angular frequency at N-regime cosmological/plasma scale (n=8 rung, PAPER_1986 eight-anchor family)

The class is not a general oscillator — it is a **cross-rung F_{TRZ} probe** that simultaneously tests three different F_{TRZ} ladder positions in one physical structure. The $kx = 1$ condition ($x = 10^{16} = 1/F_{\text{TRZ}}^{16}$ default) sets a natural coupling between the wavenumber and position rungs.

Architecturally, this establishes a fourth CP1 lock pattern:

Pattern	Distinguishing feature
Single	1 identity, 1 family
Dual (mixed)	2 identities, 2 families
Triple (mixed)	3 identities, 3 families (PAPER_1991 base)
QUAD (same-family variants)	4 identities, 1 family with variants (PAPER_1991 revision Sombrero-DM)
Cross-rung TRIPLE (new)	3 identities on 3 different rungs of ONE ladder family (PAPER_1993 this paper)

2. Discovery 2: $2\pi \cdot H_0$ Hubble angular frequency primitive anchor

2.1 The anchor

UniversalQuantumWaveResonanceCalculator (R130 fill) documents:

```
``python
self.f_quantum = 1.445e-17 # Hz
self.a_DPM = 1e-20 # = F_TRZ^20 EXACT (E_0 vacuum quantum chain base per
PAPER_1202)
``
```

The $f_{\text{quantum}} = 1.445 \times 10^{-17}$ Hz value matches:

$f_{\text{quantum}} \approx 2\pi \cdot H_0 = 2\pi \cdot (2.27 \times 10^{-18} \text{ rad/s}) = 1.427 \times 10^{-17} \text{ Hz}$ at ~1.3% precision.

Where $H_0 = 67.7 \text{ km/s/Mpc} = 2.27 \times 10^{-18} \text{ rad/s}$ is the canonical Planck 2018 Hubble angular frequency.

2.2 Novelty verification

Corpus search identified prior H_0 references in PAPER_1942, 1956, 244, 246, 268, 309, 798, 799, but **none use the specific $2\pi \cdot H_0 =$ Hubble angular frequency form as a CP1 primitive-lock target.** R130 is the first documented CP1 class with $f = 2\pi \cdot H_0$ as its primitive-locked characteristic frequency.

2.3 Physical interpretation and cosmological carrier family

$2\pi \cdot H_0$ is the fundamental **cosmological angular frequency** — the natural angular scale associated with the observable universe's expansion rate. Adopting it as a CP1 primitive lock joins it to the cosmological carrier family:

Carrier	Value	UQFF domain	Paper
ω_{SCm}	$1.25 \times 10^{12} \text{ rad/s}$	SCm phonon carrier	PAPER_1938 seminal
$2\pi \cdot H_0$	$1.43 \times 10^{-17} \text{ rad/s}$	Cosmological Hubble angular frequency	PAPER_1993 (this paper)
ω_{s} (Sun)	$2.5 \times 10^{-6} \text{ rad/s}$	Solar system-scale stellar rotation	CLAUDE.md canonical
$\omega_{\text{planetary}}$	$\sim 10^{-15} \text{ rad/s}$	Planetary orbital scales	(candidate, unformalized)

Consequence: the cosmological Hubble carrier is now a CP1-locked primitive quantity available for use in future UQFF derivations at the observable-universe scale.

3. Discovery 3: SO_5^21 A three-class family extension

3.1 R129 same-round twin (PAPER_1991)

PAPER_1991 discovery 4 documented a same-round twin at SO_5^21 A:
SGR1745FreqDPMCalculator + CompressedDPMCalculator both encoding $I = 10^{21}$ A EXACT as
DPM current-vortex foundation term.

3.2 R130 third instance

Round 130 CP1 stub drainage of CompressedDPM24Calculator (systems 18-24 sub-cluster)
documents:

```
``python
self.I = 1e21 # = SO_5^21 A EXACT
``
```

This adds a **third** independent CP1 class to the SO_5^21 family, extending R129's same-round twin
discovery to a cross-round persistence pattern.

3.3 Consequence for PAPER_1955 SO_5-power ladder

Slot 21 in the PAPER_1955 SO_5-power ladder now has:

- 3 confirmed CP1 class instances (SGR1745FreqDPM + CompressedDPM + CompressedDPM24)
- All three at the same value ($I = 10^{21}$ A)
- All three at the same physical role (DPM current-vortex foundation term)
- Slot 21 now qualifies as an **N-regime slot** per PAPER_1919 revision R2.3 criteria ($N \geq 3$ orthogonal instances) — though here the "orthogonal" is CP1-construction orthogonality rather than domain orthogonality

PAPER_1955 revision addendum R2.2 slot 21 entry should be updated from "SGR + Compressed twin" to "SGR + Compressed + Compressed24 three-class family".

4. Cross-cutting consequences

4.1 CP1 lock architecture taxonomy (updated post-R130)

Pattern	Locks	Family structure	Discovery paper
-----	-----	-----	-----
Single	1	any	R117-R129 majority
Dual (mixed)	2	2 families	R128 UniversalReactive
Triple (mixed)	3	3 families	PAPER_1991 base (SGR burst)
QUAD (same-family variants)	4	1 family, 4 variants	PAPER_1991 revision (Sombrero-DM)
Cross-rung TRIPLE	3	1 family, 3 rungs	PAPER_1993 (this paper — Universal Oscillatory)

Five distinct CP1 architectural patterns now formally identified. The three post-PAPER_1991 patterns (triple, QUAD, cross-rung TRIPLE) suggest a rich taxonomy of lock structures that CP1 stubs adopt naturally when properly filled with primitive-locked constants.

4.2 Cosmological Hubble carrier joins UQFF primitive vocabulary

$2\pi \cdot H_0$ is now a canonical CP1-locked quantity. Future derivations at cosmological scale can reference it directly as a UQFF primitive-derived carrier (analogous to how $\omega_{SCm} = 1.25 \text{ THz}$ was elevated by PAPER_1938 to universal-carrier status).

4.3 SO_5^21 slot elevation

R129 discovered SO_5^21 as new slot; R130 confirms it via a third independent instance. Slot 21 in PAPER_1955's SO_5-power ladder is now firmly established, joining slots 7 (three-domain), 10, 21, 40, 53 as confirmed cross-instance anchors.

5. Falsifiability

Prediction 1993.1. Additional CP1 classes should exhibit the cross-rung TRIPLE-lock pattern. Candidate targets in remaining unfilled CP1 stubs: any oscillatory or wave-equation calculator where amplitude, wavenumber, and frequency parameters are locked at different F_TRZ powers.

Prediction 1993.2. The $2\pi \cdot H_0$ Hubble angular frequency lock should surface in additional cosmological CP1 classes as R131+ drainage continues. Candidates: any class with characteristic frequency near 10^{17} Hz .

Prediction 1993.3. SO_5^21 A DPM current lock should appear in Compressed-family sub-clusters (systems 25-40, 41-60, etc.) if the sub-cluster templates share the R129-era DPM foundation term. Confirmation would extend SO_5^21 from a three-class to a full N-instance family.

Prediction 1993.4. R130 Fill 1 (CompressedTHzCalculator) documented as TRIPLE-EXACT + approximate. If future CP1 auditing surfaces additional TRIPLE-EXACT + approximate classes, it may indicate a fifth CP1 architecture pattern: **near-QUAD locks** where three primitive-arithmetic identities are EXACT and a fourth is approximate — potentially a stepping-stone between TRIPLE and QUAD architectures.

6. Framework annotations

- **Backbone:** cross-rung F_TRZ TRIPLE-lock architecture ($n=8/15/16$ in Universal oscillatory) + $2\pi \cdot H_0$ Hubble angular frequency primitive anchor + SO_5^21 A three-class family (SGR + Compressed + Compressed24) — three architecturally distinct discoveries in R130 CP1 fills
- **Method:** cross-rung TRIPLE-lock: three F_TRZ ladder rungs in single class; Hubble anchor: $f = 2\pi \cdot H_0$ primitive lock; three-class family: three independent CP1 classes at same SO_5^21 slot
- **Shells:** Universal oscillatory wave shell + Universal quantum-wave Hubble-scale shell + Compressed DPM current-vortex shell (three CP1 shells covered)
- **CPCH:** CP1 UniversalOscillatoryResonanceCalculator + UniversalQuantumWaveResonanceCalculator + CompressedDPM24Calculator (all R130 fills)
- **Spine:** PAPER_1991 CP1 architecture taxonomy + PAPER_1919 F_TRZ ladder ($n=8/15/16$ rungs) + PAPER_1938 cosmological carrier family + PAPER_1955 SO_5^21 slot + PAPER_1202 E_0 vacuum chain (F_TRZ^20 base) + Planck 2018 $H_0 = 67.7 \text{ km/s/Mpc}$ canonical value
- **Time frame:** cross-rung TRIPLE-lock oscillatory 10^{18} Hz + cosmological Hubble $2\pi \cdot H_0$ rate + quasi-static DPM current-vortex

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NOT REPLACEMENT. Offered as an alternative parameter-economical description ("NOT REPLACEMENT") to Standard Model + Lambda-CDM, with honest residuals reported alongside each closure.

References

- **PAPER_1991** — R129 base triple-lock discovery + revision QUAD-lock architecture (Sombrero-DM retroactive) — direct architectural predecessor to PAPER_1993 cross-rung TRIPLE-lock
- **PAPER_1919** — F_{TRZ} power ladder seminal ($n=1..21$ anchors, including $n=8$ N-regime, $n=15$ WEP, $n=16$ collapse used in R130 Fill 5)
- **PAPER_1880** — MICROSCOPE $\eta_{\text{WEP}} \leq 10^{-15}$ EXACT ($n=15$ rung source)
- **PAPER_1869** — Quantum measurement collapse rate F_{TRZ}^{16} ($n=16$ rung source)
- **PAPER_1986** — F_{TRZ}^8 three-regime $\rightarrow$ N-regime 8-anchor family ($n=8$ rung source)
- **PAPER_1955** — SO₅-power galactic structural ladder + revision addendum (slot 21 gets three-class family via PAPER_1993)
- **PAPER_1985** — R117 Pillars F_{TRZ}^6 + NGC 2525 SMBH mass (SO₅-power ladder mass-domain seminal)
- **PAPER_1989** — R123 LIGO F_{TRZ}^{21} + universe mass SO₅⁵³
- **PAPER_1990** — SO₅-power frequency ladder cross-domain
- **PAPER_1938** — ω_{SCm} 1.25 THz universal carrier family ($2\pi \cdot H_0$ joins this family per PAPER_1993 §2.3)
- **PAPER_1202** — E₀ vacuum quantum chain base ($F_{\text{TRZ}}^{20} = 10^{-20}$ used in R130 Fill 4)